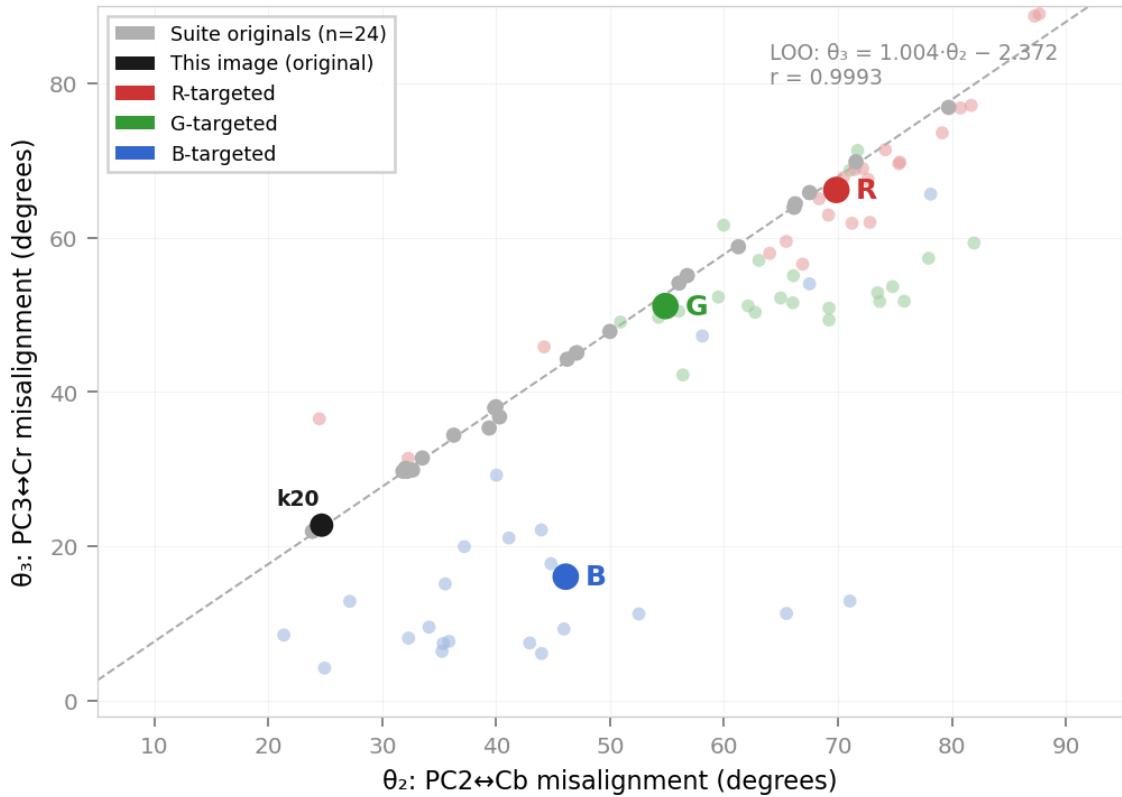


Compression Profile: KODIM20768x512 | 24-bit RGB | PNG (lossless)

P51 Mustang | Near-Degenerate (PC1 = 98.42%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9955	0.9584	0.9764	0.977	24.67°	22.74°	+0.34°
R-targeted	0.2851	0.2820	0.9940	0.520	69.86°	66.21°	-1.56°
G-targeted	0.2518	0.9842	0.2446	0.493	54.85°	51.16°	-1.54°
B-targeted	0.9438	-0.2072	-0.1820	0.444	46.1°	16.06°	-27.85°

R-targeted isolates red (R-G, R-B < 0.29) while preserving G-B coupling at 0.994.
G-targeted isolates green (R-G, G-B < 0.25) while preserving R-B at 0.984.
B-targeted preserves R-G at 0.944 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM20768x512 | 24-bit
RGB | PNG (lossless)

P51 Mustang | Near-Degenerate (PC1 = 98.42%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

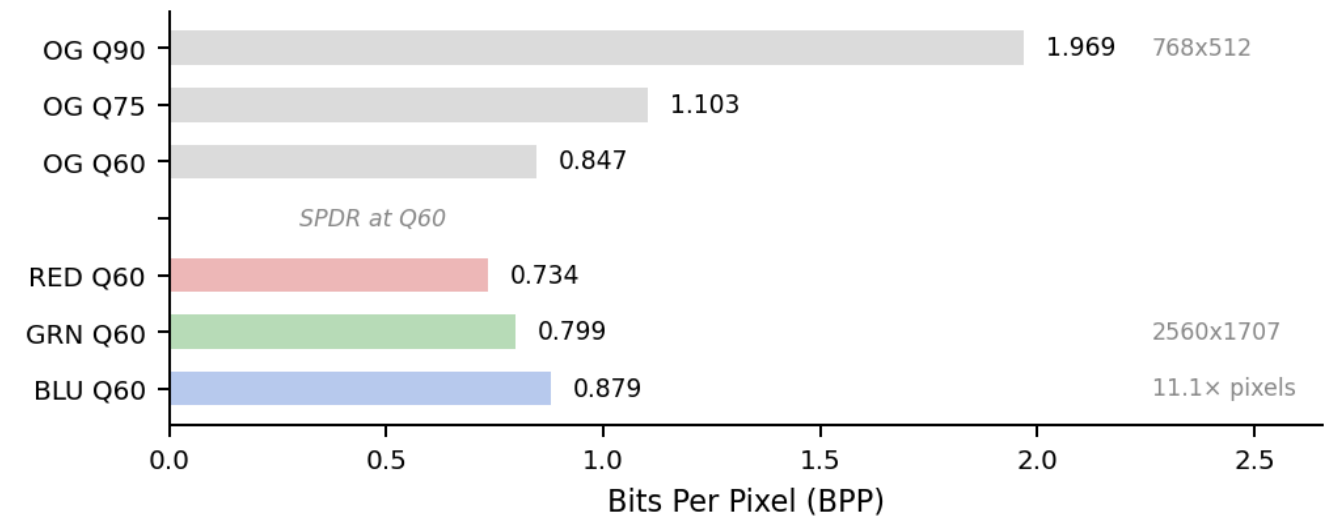
ORIGINAL (kodim20)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	480.9 KB	10.019	393,216
JPEG Q90	JPEG	768x512	94.5 KB	1.969	393,216
JPEG Q75	JPEG	768x512	52.9 KB	1.103	393,216
JPEG Q60	JPEG	768x512	40.7 KB	0.847	393,216

REDISTRIBUTED (kodim20 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	391.7 KB	0.734	4,369,920
GRN Q60	JPEG	2560x1707	426.0 KB	0.799	4,369,920
BLU Q60	JPEG	2560x1707	468.9 KB	0.879	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM20 768x512 | 24-bit RGB | PNG (lossless)

P51 Mustang | Near-Degenerate (PC1 = 98.42%)

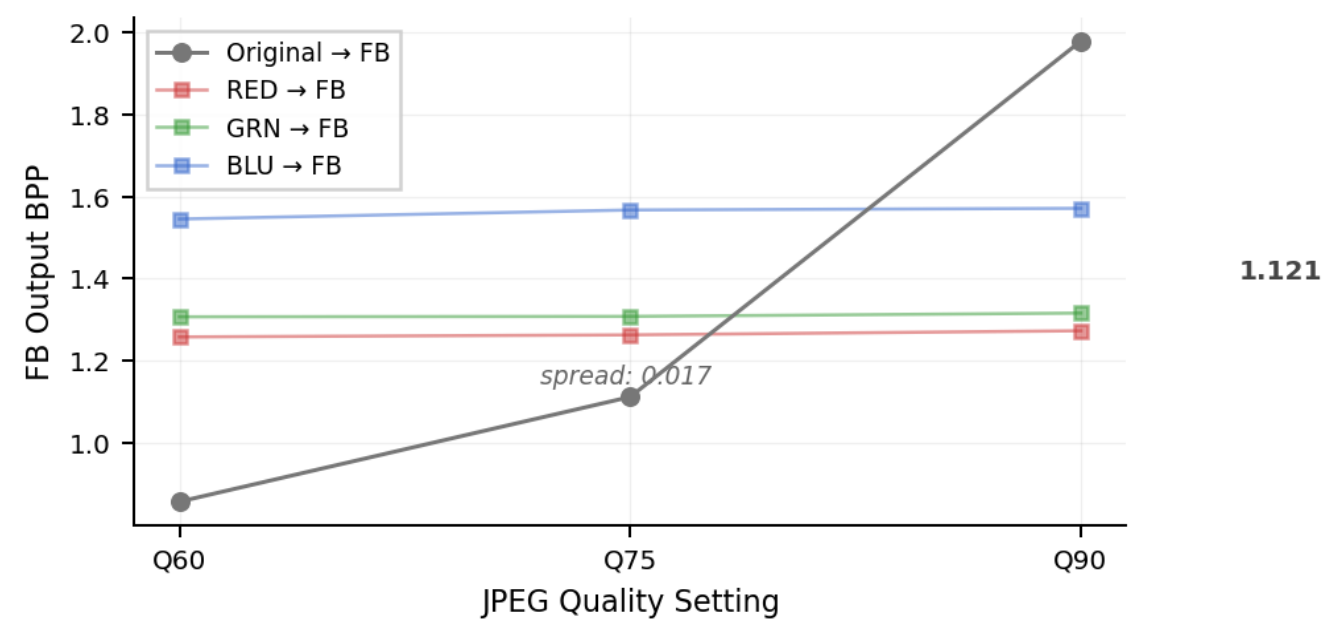
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	1.785	—
R-targeted → FB	2560x1707	2048x1366	1.273	28.7%
G-targeted → FB	2560x1707	2048x1366	1.316	26.3%
B-targeted → FB	2560x1707	2048x1366	1.571	12.0%
ABC (no perturbation) → FB	2560x1707	2048x1366	0.903	49.4%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9770 vs 0.977 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.121 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.017 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim20

Best BPP (Q60, direct)	0.734	2560x1707, 4,369,920 px
Best BPP through FB	1.273	2048x1366 output
vs OG Q90 → FB	35.6%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216